

FINITE-DIMENSIONAL SPACES WITH ZERO NUMERICAL INDEX HAVE $\mathbf{L}_{p,p}$ -NU

AGENT_LANE_19

SOURCE PROBLEM

The source paper is Sheldon Dantas, Sun Kwang Kim, Han Ju Lee, and Martin Mazzitelli, *On various types of density of numerical radius attaining operators*, arXiv:2010.00280. After proving that finite-dimensional Banach spaces with positive numerical index have $\mathbf{L}_{p,p}$ -nu, and hence that all finite-dimensional complex Banach spaces have it, the authors ask whether the remaining real zero-index finite-dimensional case also has the property:

Finite dimensional spaces with $n(X) = 0$ have the $\mathbf{L}_{p,p}$ -nu?

We answer this question affirmatively.

DEFINITIONS

Let X be a real or complex Banach space. For $T \in \mathcal{L}(X)$, its numerical radius is

$$v(T) = \sup\{|x^*(Tx)| : (x, x^*) \in \Pi(X)\},$$

where

$$\Pi(X) = \{(x, x^*) \in S_X \times S_{X^*} : x^*(x) = 1\}.$$

The space X has $\mathbf{L}_{p,p}$ -nu if for every $\varepsilon > 0$ and every state $(x, x^*) \in \Pi(X)$, there is $\eta(\varepsilon, (x, x^*)) > 0$ such that whenever $T \in \mathcal{L}(X)$ satisfies $v(T) = 1$ and

$$|x^*(Tx)| > 1 - \eta(\varepsilon, (x, x^*)),$$

there exists $S \in \mathcal{L}(X)$ such that $v(S) = 1$,

$$|x^*(Sx)| = 1, \quad \|S - T\| < \varepsilon.$$

IDEA OF THE PROOF

The positive-index proof in the source paper uses compactness of the v -unit sphere in $\mathcal{L}(X)$. When $n(X) = 0$, v is only a seminorm, so that sphere need not be bounded in the operator norm. The remedy is to remove exactly the directions invisible to numerical radius.

Let $\mathcal{Z}(X) = \{T \in \mathcal{L}(X) : v(T) = 0\}$. On the quotient $\mathcal{L}(X)/\mathcal{Z}(X)$, the numerical radius is a genuine norm. Since X is finite-dimensional, this quotient is finite-dimensional, and the v -unit sphere is compact. The state

evaluation $T \mapsto x^*(Tx)$ is constant on cosets of $\mathcal{Z}(X)$, so exact state attainment is a closed subset of that compact sphere. Compactness gives a quotient-near exact attainer; lifting that quotient perturbation gives an operator-norm-near exact attainer in $\mathcal{L}(X)$.

Lemma 1. *Let X be finite-dimensional and set*

$$\mathcal{Z}(X) = \{T \in \mathcal{L}(X) : v(T) = 0\}.$$

For a fixed $(x, x^) \in \Pi(X)$, the map*

$$\Phi_{x, x^*}(T + \mathcal{Z}(X)) = x^*(Tx)$$

is well-defined and continuous on $\mathcal{L}(X)/\mathcal{Z}(X)$. Moreover $\nu(T + \mathcal{Z}(X)) = v(T)$ is a norm on the quotient.

Proof. If $K \in \mathcal{Z}(X)$, then $v(K) = 0$, so

$$|y^*(Ky)| = 0 \quad ((y, y^*) \in \Pi(X)).$$

In particular $x^*(Kx) = 0$, proving that Φ_{x, x^*} is well-defined. It is linear, hence continuous because the quotient is finite-dimensional.

The formula $\nu(T + \mathcal{Z}(X)) = v(T)$ is well-defined by the triangle inequality for v , and its kernel is exactly the zero coset by definition of $\mathcal{Z}(X)$. Thus ν is a norm. $\square$

Theorem 2. *Every finite-dimensional real or complex Banach space has $\mathbf{L}_{p,p}$ -nu.*

Proof. Fix $\varepsilon > 0$ and $(x, x^*) \in \Pi(X)$. Let

$$Q = \mathcal{L}(X)/\mathcal{Z}(X)$$

and let $q : \mathcal{L}(X) \rightarrow Q$ be the quotient map. We use two norms on Q :

$$\nu(q(T)) = v(T), \quad \|q(T)\|_q = \inf_{K \in \mathcal{Z}(X)} \|T - K\|.$$

Since Q is finite-dimensional, these norms induce the same topology.

Consider the compact ν -unit sphere

$$A = \{u \in Q : \nu(u) = 1\}$$

and the continuous function

$$f(u) = |\Phi_{x, x^*}(u)|.$$

The closed set

$$M = \{u \in A : f(u) = 1\}$$

is nonempty: for $R_0 z = x^*(z)x$, one has $v(R_0) = 1$ and $|x^*(R_0 x)| = 1$.

We claim that there is $\eta > 0$ such that every $u \in A$ with $f(u) > 1 - \eta$ satisfies

$$\text{dist}_q(u, M) < \varepsilon,$$

where the distance is computed using the quotient operator norm $\|\cdot\|_q$. If not, there are $u_n \in A$ with $f(u_n) \rightarrow 1$ and $\text{dist}_q(u_n, M) \geq \varepsilon$. By compactness

of A , pass to a subsequence $u_n \rightarrow u_0 \in A$. Continuity gives $f(u_0) = 1$, hence $u_0 \in M$, contradicting the distance lower bound.

Now let $T \in \mathcal{L}(X)$ satisfy $v(T) = 1$ and

$$|x^*(Tx)| > 1 - \eta.$$

Then $q(T) \in A$ and $f(q(T)) > 1 - \eta$, so choose $m = q(R) \in M$ and $K \in \mathcal{Z}(X)$ such that

$$\|T - (R + K)\| < \varepsilon.$$

Put $S = R + K$. Since $q(S) = m \in M$, we have

$$v(S) = \nu(m) = 1, \quad |x^*(Sx)| = f(m) = 1,$$

and the displayed inequality gives $\|S - T\| < \varepsilon$. This is exactly $\mathbf{L}_{p,p}$ -nu. $\square$

CONSEQUENCES AND SCOPE

The source paper already proved the theorem when $n(X) > 0$. Theorem 2 removes the positivity assumption and answers the stated finite-dimensional $n(X) = 0$ question affirmatively. It does not address the other open questions in the source paper, such as whether $\mathbf{L}_{p,p}$ -nu implies strong subdifferentiability in the infinite dimensional case, or whether $\mathbf{L}_{o,o}$ -nu forces finite dimensionality without additional assumptions.

NOVELTY CHECK

I checked the run indexes for arXiv:2010.00280 and for the phrases $\mathbf{L}_{p,p}$ -nu, finite-dimensional $n(X) = 0$, zero numerical radius, and quotient by skew-hermitian operators. No duplicate packet appeared. A web search for the exact finite-dimensional zero-index question and close variants found the source arXiv page and no separate later answer. The proof uses only the quotient by zero numerical-radius operators already introduced in the source paper.

REFERENCES

- [1] S. Dantas, S. K. Kim, H. J. Lee, and M. Mazzitelli, *On various types of density of numerical radius attaining operators*, arXiv:2010.00280.